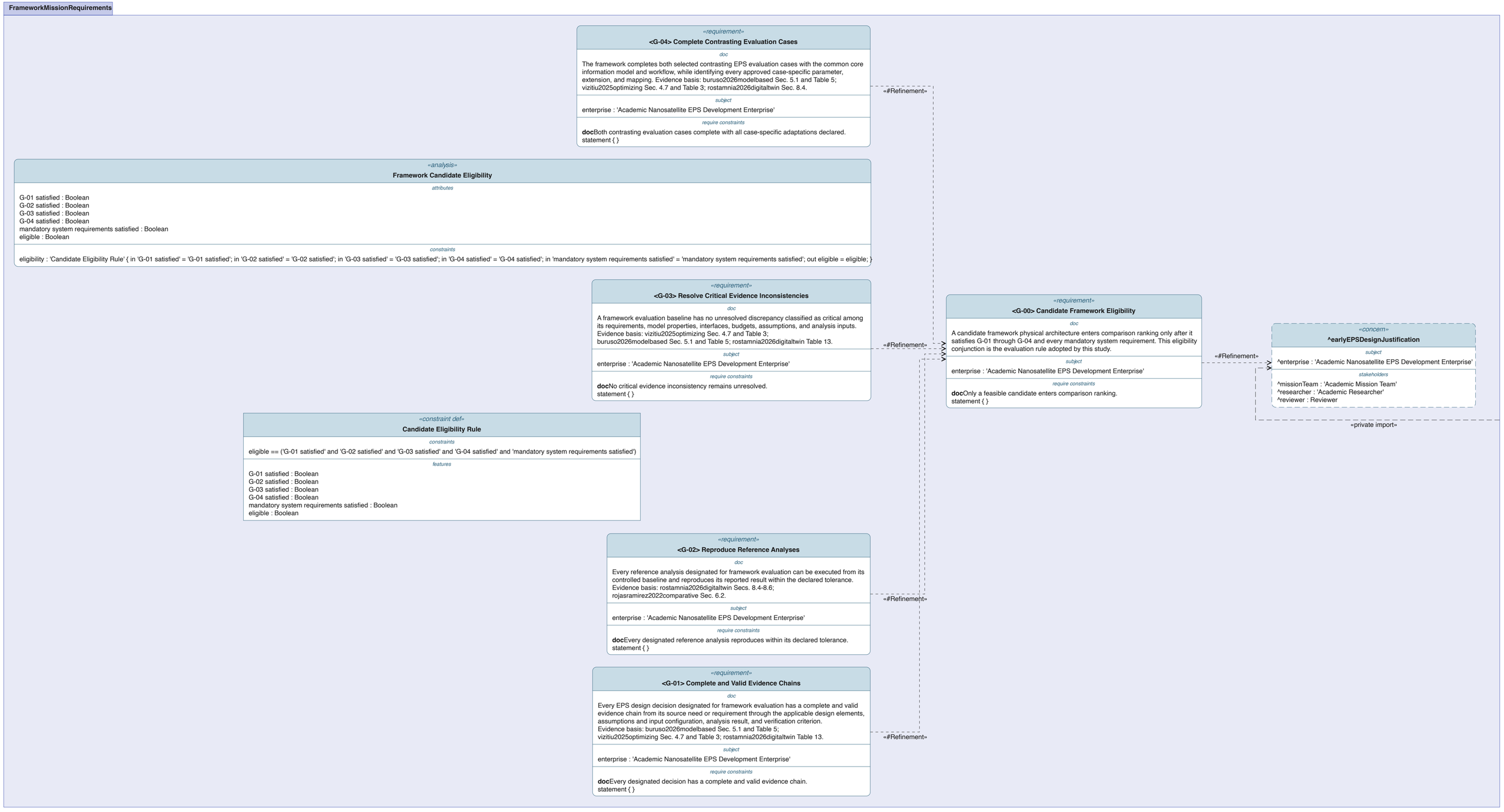
FrameworkMissionRequirements
«requirement»
<G-04> Complete Contrasting Evaluation Cases
doc
The framework completes both selected contrasting EPS evaluation cases with the common core information model and workflow, while identifying every approved case-specific parameter, extension, and mapping. Evidence basis: buruso2026modelbased Sec. 5.1 and Table 5; vizitiu2025optimizing Sec. 4.7 and Table 3; rostamnia2026digitaltwin Sec. 8.4.
subject
enterprise : 'Academic Nanosatellite EPS Development Enterprise'
require constraints
docBoth contrasting evaluation cases complete with all case-specific adaptations declared.
statement { }
«#Refinement»
«analysis»
Framework Candidate Eligibility
attributes
G-01 satisfied : Boolean
G-02 satisfied : Boolean
G-03 satisfied : Boolean
G-04 satisfied : Boolean
mandatory system requirements satisfied : Boolean
eligible : Boolean
constraints
eligibility : 'Candidate Eligibility Rule' { in 'G-01 satisfied' = 'G-01 satisfied'; in 'G-02 satisfied' = 'G-02 satisfied'; in 'G-03 satisfied' = 'G-03 satisfied'; in 'G-04 satisfied' = 'G-04 satisfied'; in 'mandatory system requirements satisfied' = 'mandatory system requirements satisfied'; out eligible = eligible; }
«requirement»
<G-03> Resolve Critical Evidence Inconsistencies
doc
A framework evaluation baseline has no unresolved discrepancy classified as critical among its requirements, model properties, interfaces, budgets, assumptions, and analysis inputs. Evidence basis: vizitiu2025optimizing Sec. 4.7 and Table 3; buruso2026modelbased Sec. 5.1 and Table 5; rostamnia2026digitaltwin Table 13.
subject
enterprise : 'Academic Nanosatellite EPS Development Enterprise'
require constraints
docNo critical evidence inconsistency remains unresolved.
statement { }
«#Refinement»
«requirement»
<G-00> Candidate Framework Eligibility
doc
A candidate framework physical architecture enters comparison ranking only after it satisfies G-01 through G-04 and every mandatory system requirement. This eligibility conjunction is the evaluation rule adopted by this study.
subject
enterprise : 'Academic Nanosatellite EPS Development Enterprise'
require constraints
docOnly a feasible candidate enters comparison ranking.
statement { }
«#Refinement»
«concern»
^earlyEPSDesignJustification
subject
^enterprise : 'Academic Nanosatellite EPS Development Enterprise'
stakeholders
^missionTeam : 'Academic Mission Team'
^researcher : 'Academic Researcher'
^reviewer : Reviewer
«private import»
«constraint def»
Candidate Eligibility Rule
constraints
eligible == ('G-01 satisfied' and 'G-02 satisfied' and 'G-03 satisfied' and 'G-04 satisfied' and 'mandatory system requirements satisfied')
features
G-01 satisfied : Boolean
G-02 satisfied : Boolean
G-03 satisfied : Boolean
G-04 satisfied : Boolean
mandatory system requirements satisfied : Boolean
eligible : Boolean
«requirement»
<G-02> Reproduce Reference Analyses
doc
Every reference analysis designated for framework evaluation can be executed from its controlled baseline and reproduces its reported result within the declared tolerance. Evidence basis: rostamnia2026digitaltwin Secs. 8.4-8.6; rojasramirez2022comparative Sec. 6.2.
subject
enterprise : 'Academic Nanosatellite EPS Development Enterprise'
require constraints
docEvery designated reference analysis reproduces within its declared tolerance.
statement { }
«#Refinement»
«requirement»
<G-01> Complete and Valid Evidence Chains
doc
Every EPS design decision designated for framework evaluation has a complete and valid evidence chain from its source need or requirement through the applicable design elements, assumptions and input configuration, analysis result, and verification criterion. Evidence basis: buruso2026modelbased Sec. 5.1 and Table 5; vizitiu2025optimizing Sec. 4.7 and Table 3; rostamnia2026digitaltwin Table 13.
subject
enterprise : 'Academic Nanosatellite EPS Development Enterprise'
require constraints
docEvery designated decision has a complete and valid evidence chain.
statement { }
«#Refinement»